

1. AudioRemove-AC3.ps1 is a pure-remux utility.

No video re-encoding; only remuxing streams.

Purpose:

Remove all AC3 and E-AC3 audio streams (typically are **low bit-rate**)

Result:

"Lean Master" movie contains only video and high-fidelity, which is retained (TrueHD/AAC 7.1/DTS-HD MA).

Use:

Prepares **mkv** movie for Conversion Engines (DDP51 / Keep71) to generate high-quality, high-bitrate (1024k) E-AC3 5.1 compatibility tracks.

2. Requirements

To run AudioRemove-AC3.ps1, you need:

- PowerShell 7.6
- FFmpeg 8.1 (ffmpeg + ffprobe must be placed next to the script)

Cross-Platform Compatibility

- Windows 11
- Linux (Ubuntu, Debian, Arch, Fedora)
- macOS (Ventura, Sonoma)

As long as PowerShell 7.6 is installed and FFmpeg 8.1 binaries are placed next to the script, it works identically across all platforms.

Official FFmpeg download page: <https://ffmpeg.org/download.html>

Quick Start

1. Place `AudioRemove-AC3.ps1`, `ffmpeg.exe`, and `ffprobe.exe` in the same folder.
2. Run the script:

```
pwsh -ExecutionPolicy Bypass -File .\AudioRemove-AC3.ps1 ".\YourMovie.mkv"
```

i The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

- The script will create a new file: YourMovie_**remux**.mkv
- If the file exists, it will auto-increment safely.

3. Supported Codecs

Audio Codecs — Kept

Codec Family	Examples
TrueHD	TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos
DTS	DTS Core, DTS-HD MA, DTS-HD HRA
FLAC	All FLAC channel layouts
AAC	AAC LC, HE-AAC, AAC 5.1, AAC 7.1
Opus	Stereo + multichannel
PCM	<i>pcms16le</i> , <i>pcms24le</i> , <i>pcm_f32le</i>

Audio Codecs — Removed

Codec Family	Examples
AC3	<i>ac3</i>
E-AC3	<i>eac3</i>

Subtitle Streams

- English Subtitles Kept automatically.
- Untagged Subtitles Kept with warning:
“Subtitle stream(s) with no language tag found — keeping as language is unknown.”
- Subtitles with a non-eng language tag are dropped silently.

Attachments — Kept

- Fonts
- Cover art
- Any MKV attachment stream

Data Streams — Dropped

- Removed with warning
- Typically unused by players

4. Global FFmpeg Options

Ensure stable remuxing, correct metadata handling, and safe processing of large files.

`-probesize 200M` & `-analyzeduration 200M`

What: Deep probe of the input container.

Why: Ensures accurate detection of audio/subtitle streams, especially in large MKVs.

`-max_muxing_queue_size`

What: Prevents video flooding mux queue on long files.

Value: 14000 (1024-100000) recommended range

`-fflags +discardcorrupt`

What: Discards corrupt packets instead of aborting.

Why: Prevents remux failures on slightly damaged files.

`-avoid_negative_ts make_zero`

What: Normalizes timestamps to start at zero.

Why: Prevents sync drift caused by negative PTS values.

`-map_metadata 0` & `-map_chapters 0`

What: Copies global metadata and chapter markers.

Why: Preserves the original file's structure.

`-map 0:v?`

What: Maps all video streams (excluding attached pictures unless explicitly present).

Why: Ensures video is preserved without modification.

`-c copy`

What: Stream-copy mode for all mapped streams.

Why: Guarantees zero quality loss and maximum speed.

Playback Assignment Rules

To ensure smooth playback, the rules are as follows:

- The first kept audio track will be set as **default**
- All other audio tracks will be set to **none**

5. Summary Report

After processing, the script generates a detailed table:

Column	Description
Idx	Stream's original number. For dropped-subtitle row, we use a placeholder instead.
Codec	Codec used for audio, subtitle, or attachment
Action	Whether the stream was kept or removed
Rule	Reason for keeping or removing the stream

Color Coding for above table

- **Red** > Removed AC3/E-AC3
- **Green** > Kept audio
- **Cyan** > Kept subtitles
- **Magenta** > Kept attachments
- **Yellow** > Non-eng Subtitles dropped

6. Non-Critical FFmpeg Warnings (Safe to Ignore)

These warnings may appear during remuxing but **do not affect output quality**.

“discarding packet”

- Caused by minor corruption in the source
- Safe due to `+discardcorrupt`

“non-monotonous DTS”

- Happens when timestamps overlap
- Remux corrects this automatically

“Subtitle: unspecified size”

- Common for PGS subtitles
- Subtitles are copied untouched

“Estimating duration from bitrate, this may be inaccurate”

- FFmpeg recalculates duration during muxing
- Safe to Ignore